How a car works

A car is a collection of systems that generate power in an engine and transmit it to the wheels. Other systems allow the driver to control the car by turning the wheels to change direction and by applying brake force to either slow down or stop.

WHAT WAS THE FIRST MASS-PRODUCED CAR WITH AN AUTOMATIC TRANSMISSION?

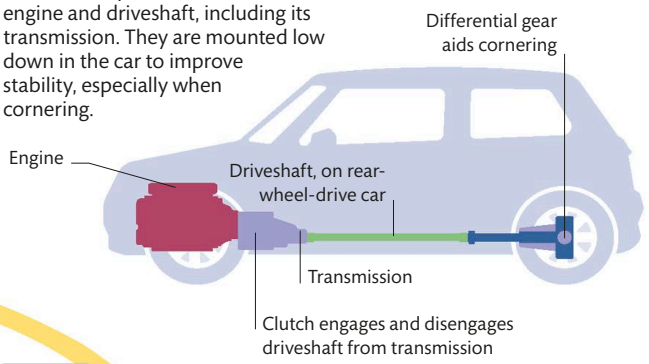
The first fully automatic transmission was an optional extra on Oldsmobile cars in the US from 1940.

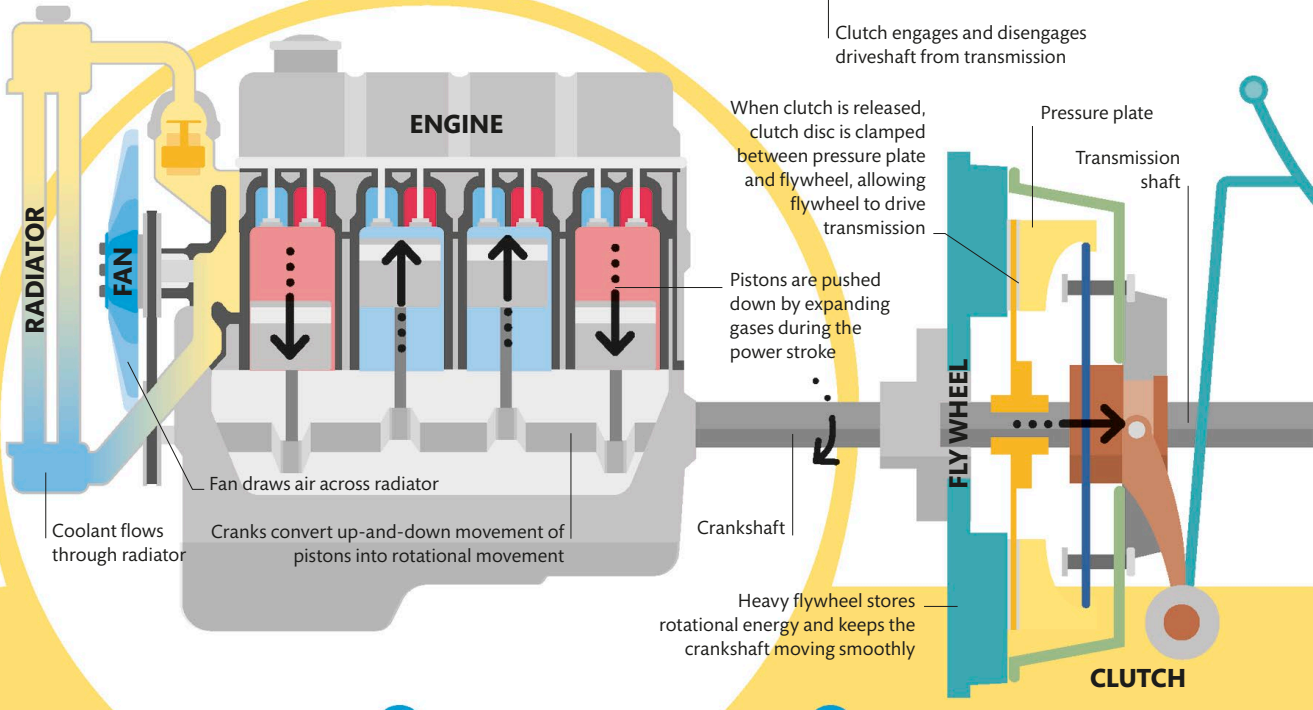
Transmitting power

A car's engine is linked to its wheels by a system of shafts and gears, collectively known as the drivetrain, which make use of the engine power in the most efficient way. Most cars have a two-wheel-drive arrangement, in which either the two front wheels or two back wheels are driven by the engine. Off-road vehicles, which require more grip on unstable surfaces, have four-wheel drive, meaning all four wheels are driven directly by the engine.

Inside the car

The heaviest parts of a car are its engine and driveshaft, including its transmission. They are mounted low down in the car to improve stability, especially when cornering.





Driving off

A car is set in motion by a series of operations that generate power and transfer it to the driven wheels in a controlled way. Turning the ignition key or pressing the start button switches on a small battery-powered electric motor, which starts the car's piston engine.

1 Engine

A car's motion begins with its engine. Starting the engine ignites fuel and releases energy (see pp.42–43). This moves the pistons, which turn the engine's crankshaft. A flywheel attached to the crankshaft smooths out the power provided by the pistons.

2 Clutch

In a car with a manual transmission, when the car first starts, the driver must push the clutch pedal in to disconnect the engine from the wheels so that the car does not lurch forward. The driver then releases the clutch pedal, allowing the engine to turn the wheels.